Lecture 9

DiD, Synthetic Control & ML Integration

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	ML CONNECTION
1	TWFE Crisis	Why standard DiD fails with staggered adoption
2	Modern DiD	CS2021 / SA / BJS — clean comparison groups
3	Synthetic Control	SCM as constrained ML optimization
4	DML for DiD	ML for high-dimensional controls
5	RDD + ML	Covariate adjustment without bias
6	Unified Framework	When to use what, and how tools combine

Guiding question: Three tools, one goal — how do DiD, SC, and ML reinforce each other?

Part 1

The TWFE Crisis

The Problem: Staggered Adoption

Standard DiD (2×2): Two groups, two periods — clean.

Real world: Treatment rolls out at *different times* across units.

UNIT	TREATED?	WHEN?	TWFE DOES WHAT?
Early adopters	Yes	Period 2	Uses as **controls** for later cohorts
Late adopters	Yes	Period 6	Compared against early adopters
Never treated	No	—	Clean controls

The forbidden comparison: Late vs Early at $t = 6$ estimates $ATT_{\text{Late}} - ATT_{\text{Early}}$, not a treatment effect. If early adopters' effects *grow over time*, this comparison can be **negative** even when the true effect is positive.

How Bad Can It Get?

Goodman-Bacon (2021): TWFE = weighted average of all 2×2 DiDs.

Some of those 2×2 DiDs use **already-treated units as controls**.

$$\hat{\beta}^{TWFE} = \underbrace{w_1\delta_1 + w_2\delta_2 + w_3\delta_3}_{\text{valid comparisons}} + \underbrace{w_4\delta_4^{\text{neg}}}_{\text{forbidden}} < \text{True ATT}$$

(Schematic — Goodman-Bacon 2021, Figure 1: in the NLSY job displacement application, TWFE attenuates the true effect by mixing in negative-weight comparisons.)

Downward bias or sign reversal — when treatment effects are heterogeneous and dynamic.

> *"TWFE is unbiased only if treatment effects are constant and homogeneous across cohorts and time. That's rarely true.*

What We Need: Three Separate Problems

PROBLEM	CAUSE	FIX
Invalid comparisons	Staggered adoption + heterogeneous effects	CS2021 / SA2021 / BJS
Parallel trends fails	No comparable control units	Synthetic Control / SyDiD
High-dim confounders	Many covariates, nonlinear selection	DML-DiD

These are complementary, not competing. A typical modern paper uses all three.

Part 2

Modern DiD: Clean Comparisons

Callaway & Sant'Anna (2021): The Core Idea

Instead of one $\hat{\beta}$, estimate each cohort-time effect separately:

$$ATT(g,t) = E[Y_t - Y_{g-1} \mid G_g = 1] - E[Y_t - Y_{g-1} \mid C = 1]$$

- g = cohort (first treatment period)
- t = calendar time
- C = clean control group (never-treated or not-yet-treated)

Then aggregate however you want — overall ATT, event study, cohort-specific.

Key restriction: Never use already-treated units as controls. Period.

Event Study from CS2021

Aggregate to relative event time $e = t - g$:

$$\theta_{ES}(e) = \sum_g w_g \cdot ATT(g, g+e)$$

w_g = cohort-size weights. Pre-trends ($e < 0$) should be flat — testable assumption.

Green subsidies example (staggered 2017–2020 rollout):

e	$\hat{\theta}_{ES}(e)$	SIGNIFICANT?
-2, -1	≈ 0	No — pre-trend flat ✓
0, 1	near 0	No — no immediate effect
2+	positive, growing	Yes — delayed effect

(Pattern from Callaway & Sant'Anna 2021, Figure 4: event-study estimates in the job displacement application show pre-trend validity and post-event dynamics.)

Interpretation: Pre-trends flat confirms parallel trends; delayed positive effects are consistent with an investment-to-outcome pipeline.

CS2021: Doubly Robust Estimation

Three ways to estimate ATT(g,t) — CS2021 uses the best:

Outcome Regression (OR): Model $\hat{E}[\Delta Y|D=0,X]$, predict control counterfactual.

IPW: Reweight controls by $\hat{p}_g(X)=P(G_g=1|X)$.

Doubly Robust (DR) — combine both:

$$\widehat{ATT}^{DR}(g,t)=\frac{1}{n_g}\sum_{i\in g}[\Delta Y_i-\hat{m}(X_i)] \\ -\frac{1}{n_C}\sum_{i\in C}\frac{\hat{p}_g(X_i)}{1-\hat{p}_g(X_i)}[\Delta Y_i-\hat{m}(X_i)]$$

where $\Delta Y_i=Y_{it}-Y_{i,g-1}$ and $\hat{m}(X_i)=\hat{E}[\Delta Y|D=0,X_i]$.

Why doubly robust? Consistent if *either* $\hat{m}$ or $\hat{p}_g$ is correctly specified — not both required.

ML connection: Swap logit/OLS for any ML learner → that's the DML-DiD extension in Part 4.

SA2021 and BJS: Two Variants

	CS2021	SUN-ABRAHAM (SA)	BORUSYAK-JARAVEL-SPIESS (BJS)
Estimand	$ATT(g,t)$ per cohort-time	$ATT(e)$ per relative time	Efficient overall ATT
Controls	Never or not-yet-treated	Never-treated only	All untreated obs
Main use	Flexible aggregation	Clean event study	Efficiency gains
Limitation	Many estimates to examine	Effects depend only on e, not t	Sensitive to $Y(0)$ model

Practical advice: Start with CS2021 for transparency. Check with SA or BJS for robustness. All three should tell the same story if your design is clean.

Implementation: `Lecture09-Exercise.ipynb` — Parts 1 & 2 walk through CS2021, SA, and BJS on simulated staggered data.

Part 3

Synthetic Control as ML Optimization

When Parallel Trends Fails

CS2021 still requires: Even the *weighted* control group follows parallel trends.

What if no control unit is comparable?

- California passes Proposition 99 (tobacco tax) — no other state is similar
- A country adopts a policy — no natural control group exists
- Pre-trends in raw data look non-parallel

Synthetic Control (Abadie, Diamond, Hainmueller 2010):
Don't assume parallel trends. Instead, *construct* a synthetic control that matches the treated unit before treatment.

SCM as a ML Problem

The optimization:

$$\min_w \sum_{t \leq T_0} \left(Y_{1t} - \sum_{j \geq 2} w_j Y_{jt} \right)^2$$
$$\text{s.t. } w_j \geq 0, \quad \textstyle\sum_j w_j = 1$$

Constrained least squares — solvable via quadratic programming directly.

Why the constraints matter:

- $w_j \geq 0$: No negative weights (unlike TWFE!)
- $\sum_j w_j = 1$: Extrapolation-safe

The estimated effect:

$$\hat{\tau}_t = Y_{1t} - \sum_j \hat{w}_j Y_{jt}, \quad t > T_0$$

Synthetic DiD (Arkhangelsky et al. 2021): Combining Both

Problem with pure SCM: Only one treated unit; effects estimated one time period at a time.

Problem with pure DiD: Requires parallel trends assumption to hold exactly.

SyDiD combines:

1. **SC weights $\hat{w}_j$** → select the best control units (data-driven)
2. **DiD differencing** → remove any remaining level differences

$$\hat{\tau}^{SyDiD} = \underbrace{(\bar{Y}_{1,\text{post}} - \bar{Y}_{1,\text{pre}})}_{\text{treated } \Delta} - \underbrace{\sum_j \hat{w}_j(\bar{Y}_{j,\text{post}} - \bar{Y}_{j,\text{pre}})}_{\text{weighted control } \Delta}$$

Key insight: SyDiD = DiD with a ML-selected control group. Handles multiple treated units and staggered adoption.

Implementation: `Lecture09-Exercise.ipynb` — Part 3 implements SCM and SyDiD on the green subsidies panel.

SyDiD: Two Types of Weights

Unit weights $\hat{\omega}$: find control weights that minimize pre-treatment tracking error, with ridge penalty:

$$\hat{\omega} = \arg\min_{\omega \in \Delta^{N_0}} \sum_{t \leq T_0} \left(\bar{Y}_{1t} - \sum_j \omega_j Y_{jt} \right)^2 + \zeta^2 N_0 \|\omega\|^2$$

Time weights $\hat{\lambda}$: upweight pre-periods that look most like the post-treatment period:

$$\hat{\lambda} = \arg\min_{\lambda \in \Delta^{T_0}} \sum_j \left(\bar{Y}_{j,\text{post}} - \sum_{t \leq T_0} \lambda_t Y_{jt} \right)^2$$

Why Ridge Penalty Matters

Classical SCM risk: When T_0 is small, exact pre-treatment fit can be spurious — overfitting to noise.

SyDiD solution: Ridge penalty $\zeta^2 N_0 ||\omega||^2$ shrinks weights toward uniform, improving out-of-sample validity.

ML interpretation: Unit weight selection = ridge-regularized constrained regression. Both unit and time weight problems are convex QP — fast and scalable.

DiD vs SCM vs SyDiD: When to Use

SETTING	BEST METHOD
Many treated units, good parallel trends	**CS2021 / SA / BJS**
One treated unit, good pre-treatment fit	**SCM**
One treated unit, imperfect fit	**Synthetic DiD**
Many treated, parallel trends uncertain	**Synthetic DiD**
All of the above + many covariates	**+ DML** (next section)

Part 4

DML for DiD: ML Handles the Confounders

The Selection Problem

CS2021 and SyDiD assume: Conditional on observed X, parallel trends holds.

Challenge: Selection into treatment is complex — 200+ covariates (firm text, patent embeddings, financials) with nonlinear relationships. $p \approx n$ or $p > n$ — OLS / logit break down.

Naive fix: "Just add all controls to CS2021"

Why it fails: Logit diverges at high dimensions; linear functional form is misspecified; bias propagates directly to the treatment effect estimate.

DML-DiD (Chang 2020): Use ML for the nuisance functions, maintain valid inference.

DML-DiD: Why Naive Plug-in Fails

The plug-in approach: Estimate $\hat{g}(X)$ and $\hat{\ell}(X)$ with ML, substitute directly.

The problem: ML estimators are regularized → incur bias $\|\hat{g} - g_0\|_2 = O_p(n^{-1/4})$, not $O_p(n^{-1/2})$.

Linearize the estimating equation around true nuisances (g_0, ℓ_0):

$$\hat{\tau}_{\text{plug-in}} - \tau_0 = \underbrace{\frac{1}{\sqrt{n}} \sum_i \psi_i}_{O_p(n^{-1/2})} + \underbrace{\partial_g[\cdot] \cdot (\hat{g} - g_0)}_{O_p(n^{-1/4})} + \dots$$

The first-order ML bias dominates the $1/\sqrt{n}$ variance term — standard errors are invalid.

The fix: Design the score so $\partial_g[\cdot] = 0$ at the true nuisances (Neyman orthogonality).

The DML-DiD Framework

Two nuisance functions to estimate:

- $g(X) = P(\text{treated cohort} \mid X)$ — propensity score
- $\ell(X) = E[\Delta Y \mid \text{control}, X]$ — outcome model for controls

Two ingredients that make inference valid:

> ***Neyman Orthogonality****: The moment condition is insensitive to first-order errors in $\hat{g}$ and $\hat{\ell}$. ML estimation error (bias from regularization) cancels in the score.*

> ***Cross-Fitting****: Train nuisance models on k folds, predict out-of-sample. Prevents overfitting bias. Same idea as cross-validation.*

Result: We get $\sqrt{n}$-consistent, normal inference for the ATT — even with Random Forests or XGBoost estimating the nuisances.

Neyman Orthogonality: Definition

Definition: Score $\psi(W;\tau,\eta)$ is Neyman orthogonal if:

$$\partial_\eta \, E\big[\psi(W;\tau_0,\eta)\big]\big|_{\eta=\eta_0} = 0$$

The expected score is insensitive to first-order errors in the nuisance η.

DML-DiD score (Chang 2020):

$$\psi_i(\tau; g, \ell) = \frac{(D_i - g(X_i)) \cdot (Y_i - \ell(X_i))}{g(X_i)(1 - g(X_i))} - \tau$$

Verification: $\partial_g E[\psi] = 0$ and $\partial_\ell E[\psi] = 0$ at true nuisances (g_0, ℓ_0).

Neyman Orthogonality: Why the Score Works

Step-by-step intuition:

1. **Define components:** Let $W_i = (Y_i, D_i, X_i)$ and write:

$$\psi_i = \underbrace{\frac{D_i - g(X_i)}{g(X_i)(1 - g(X_i))}}_{\text{weight } w_i} \cdot \underbrace{(Y_i - \ell(X_i))}_{\text{residual } \varepsilon_i} - \tau$$

2. **Expectation conditional on X:** At true nuisances (g_0, ℓ_0):

$$E[\psi_i \mid X_i] = \frac{E[D_i \mid X_i] - g_0(X_i)}{g_0(1 - g_0)} \cdot (E[Y_i \mid X_i] - \ell_0(X_i)) = 0$$

3. **Derivative w.r.t. g:** Perturb $g \to g_0 + t \cdot \delta g$:

$$\partial_g E[\psi]\Big|_{g_0} = E\left[\frac{\partial}{\partial g}\left(\frac{D - g}{g(1 - g)}\right)\Bigg|_{g_0} \cdot (Y - \ell_0)\right] = 0$$

4. **Derivative w.r.t. ℓ:** Perturb $\ell \to \ell_0 + t \cdot \delta \ell$:

$$\partial_\ell E[\psi]\Big|_{\ell_0} = -E\left[\frac{D - g_0(X)}{g_0(1 - g_0)}\right] = 0$$

Consequence: ML estimation error contributes only **second-order** bias → $O_p(n^{-1/2})$ inference is valid even when ML rate is

Cross-Fitting: The Algorithm

Problem: Training nuisances on the *same data* used to form ψ_i introduces overfitting correlation.

Solution: K-fold cross-fitting

1. Split sample randomly into K folds: $I_1, \dots, I_K$
2. For each fold k:
   - Train $\hat{g}^{(-k)}$ and $\hat{\ell}^{(-k)}$ on **all folds except** I_k
   - Compute scores $\hat{\psi}_i$ for $i \in I_k$ using out-of-fold predictions
3. Combine scores across all folds: $\hat{\tau}_{DML} = \frac{1}{n} \sum_k \sum_{i \in I_k} \hat{\psi}_i$

Why this works: $\hat{g}^{(-k)}$ is independent of (Y_i, D_i) for $i \in I_k$ — no overfitting correlation in the score.

Analogy: Same fold structure as cross-validation, but the goal is bias removal, not model selection.

DML-DiD vs Standard CS2021: What Changes

Chang (2020, *Econometrics Journal*) applies DML-DiD to the Sequeira (2016) tariff-and-corruption data:

METHOD	WHAT IT CONTROLS	KEY CHANGE VS NAIVE
TWFE	Linear state FE	Baseline — biased with staggering
CS2021 (logit)	Logit propensity score	Correct comparisons; removes staggering bias
CS2021 + DML	**RF/LASSO nuisances**	**Robust to nonlinear selection on observables**

When does DML change the answer? When confounders (demographics, industry mix) interact nonlinearly with treatment assignment. DML uses ML to absorb this — standard logit cannot.

Validation: If CS2021 (logit) and CS2021 + DML give similar estimates, selection on observables is likely linear and standard controls suffice.

Implementation: `Lecture09-Exercise.ipynb` — Part 4 runs DML-DiD with Random Forest nuisances and compares to linear CS2021.

Part 5

RDD + ML: Precision Without Bias

RDD Basics

Sharp RDD: Treatment $D_i = \mathbf{1}(X_i \geq c)$ determined by running variable.

Target: Local Average Treatment Effect at the cutoff

$$\tau = \lim_{x \downarrow c} E[Y \mid X = x] - \lim_{x \uparrow c} E[Y \mid X = x]$$

Identification: Units just above and below cutoff are locally randomized — only a discontinuity in Y at c is causal.

The power problem: RDD uses only data near the cutoff.

- Bandwidth h trades off bias (wider = more) vs variance (wider = less)
- With rich covariates Z, traditional approach adds them linearly — but this is often misspecified

Bandwidth Selection and the Role of ML

MSE-optimal bandwidth balances squared bias against variance:

$$h^* \propto \left(\frac{\sigma^2}{n \cdot B^2} \right)^{1/5}$$

where B = regression curvature at cutoff (bias), σ^2 = residual variance (variance).

How ML covariate adjustment changes the tradeoff:

If covariates Z explain 40% of Y variance → $\hat{\sigma}^2$ shrinks by 40%

$$h^*_{\text{ML}} \approx (0.6)^{1/5} \cdot h^*_{\text{no ML}} \approx 0.90\, h^*_{\text{no ML}}$$

Practical implication:

- Smaller bandwidth → fewer observations, but less contamination from far-from-cutoff units
- Same confidence interval width despite smaller window
- A genuine precision gain with no identification cost

ML-RDD: Nonparametric FWL

Classical FWL Theorem: In linear regression $Y = \tau D + Z'\beta + \varepsilon$:

$$\hat{\tau}_{OLS} = (\tilde{D}'\tilde{D})^{-1}\tilde{D}'\tilde{Y}, \quad \tilde{Y} = M_Z Y,\ \tilde{D} = M_Z D$$

Residualizing Z gives the same $\hat{\tau}$ as the full regression.

Nonparametric FWL (Noack, Olma, Rothe 2024): Replace M_Z with ML:

1. $\tilde{Y}_i = Y_i - \hat{E}[Y \mid Z_i]$ using any ML learner (trained on one fold)
2. Apply standard local polynomial RDD to $(\tilde{Y}_i, X_i)$ on held-out fold
3. Cross-fit: swap folds and average

Guarantee: As long as the ML learner converges at *any* rate, the RDD estimate remains consistent and inference is valid — the nonparametric FWL theorem holds.

ML-RDD (Noack, Olma, Rothe 2024): The Same DML Idea

Insight: The DML logic from Part 4 applies directly to RDD.

Procedure (RDFlex):

1. **Partial out covariates** using ML: $\tilde{Y}_i = Y_i - \hat{E}[Y \mid Z_i]$ (trained on one fold)
2. **Apply standard RDD** to residuals $\tilde{Y}$ (evaluated on held-out fold)
3. **Swap folds and average** (same cross-fitting as DML-DiD)

Why this works:

- Covariate adjustment reduces residual variance → tighter SEs
- Using ML instead of linear OLS captures non-linear relationships
- Cross-fitting prevents overfitting → no bias

The "free lunch": Variance reduction at zero cost to identification validity.

Implementation: `Lecture09-Exercise.ipynb` — Part 5 runs standard RDD vs ML-RDD side by side.

Part 6

Unified Framework: When to Use What

How the Three Tools Fit Together

One goal: Estimate the causal effect of a policy that was not randomly assigned.

Three distinct identification challenges — each needs a different tool:

CHALLENGE	ROOT CAUSE	FIX
Invalid control groups	Staggered adoption, already-treated controls	CS2021 / SA / BJS
No comparable controls	Parallel trends fails in raw data	SCM / SyDiD
High-dim selection	Many covariates, nonlinear treatment	DML-DiD / ML-RDD

Stacking the Tools

They stack: Start with CS2021 for clean comparisons, add SyDiD if parallel trends is uncertain, layer in DML if covariates are rich.

> *Typical modern paper: CS2021 → SyDiD → DML-DiD → ML-RDD (complementary design)*

Decision Guide

Step 1: How many treated units?

- One → **SCM** (if good pre-fit) or **SyDiD** (if not)
- Many → **CS2021** or **SyDiD**

Step 2: Does parallel trends hold?

- Yes (clean pre-trends) → **CS2021**
- Uncertain → **SyDiD** (data-driven control weights)

Decision Guide (Continued)

Step 3: What's the covariate situation?

- Few, linear → standard CS2021 with logit/OLS controls
- Many, nonlinear → **DML-DiD** (wrap CS2021 with ML nuisances)

Step 4: Is there a sharp cutoff?

- Yes → also run **RDD / ML-RDD** as a complementary design

Example: Green subsidies face all three challenges → CS2021 + DML-DiD + ML-RDD

Method Comparison: What Each Tool Adds

When the same policy is analyzed by all methods (e.g., Callaway & Sant'Anna 2021, Arkhangelsky et al. 2021):

METHOD	BIAS SOURCE ADDRESSED	WHAT CHANGES VS TWFE
TWFE	None	Baseline — sign/magnitude unreliable with staggering
CS2021	Forbidden comparisons	Correct cohort-time effects; reveals lag structure
SyDiD	Parallel trends + scale	Reweights time and units; robust to trend divergence
DML-DiD	Nonlinear covariate selection	Removes high-dim confounders ML-style
ML-RDD	Cutoff assignment	Identifies effect at threshold; sharp design

> *The diagnostic: if TWFE and CS2021 give very different signs or magnitudes, staggering contamination is present. Use CS2021 or SyDiD as the baseline.*

Key Takeaways

TOOL	SOLVES	ML ROLE
CS2021 / SA / BJS	Forbidden comparisons in staggered DiD	Optional (DML extension)
SCM	No comparable controls	Constrained optimization = ML
SyDiD	Parallel trends + scale	ML selects control weights
DML-DiD	High-dim confounders in DiD	Core: RF/XGBoost for nuisances
ML-RDD	Low power at cutoff	Core: GBM for covariate partial-out

The common thread: ML handles the hard parts — nuisance estimation, weight selection, covariate adjustment — while the econometric identification logic (parallel trends, local randomization) remains in charge.

Part 7

Robustness & Placebo Tests

Why Placebo Tests Matter

The core question: Is the estimated effect real, or an artifact of the research design?

Three common placebo strategies:

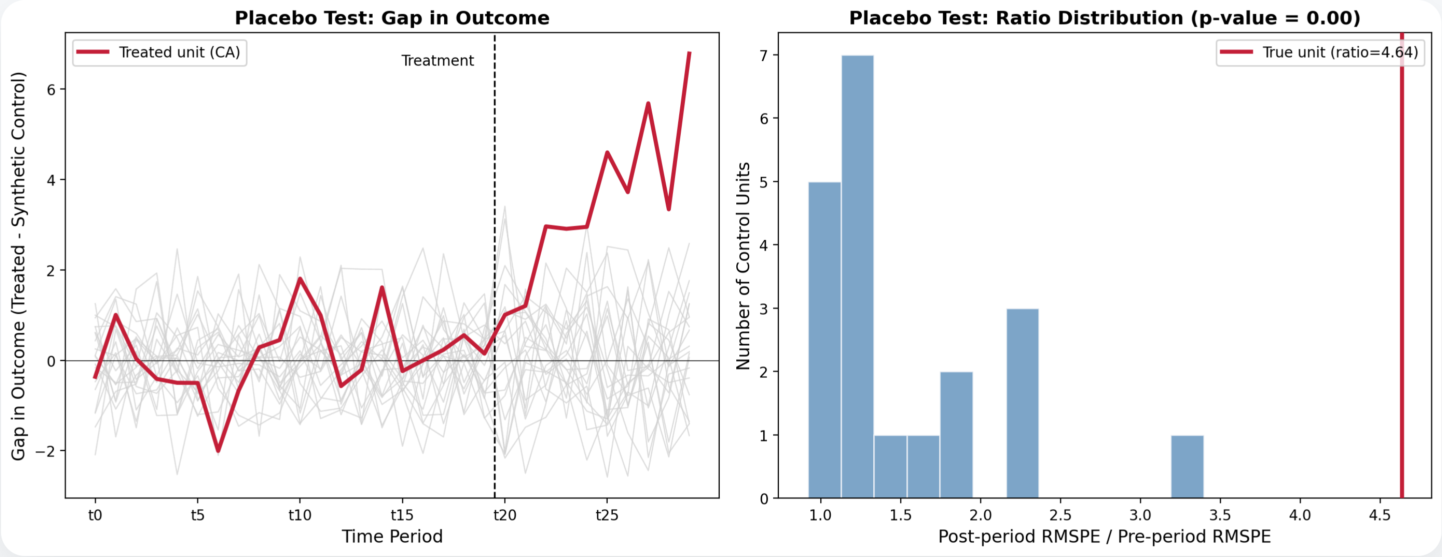
STRATEGY	WHAT YOU DO	WHAT IT TESTS
Time placebo	Pretend treatment happened earlier	Pre-trends should be flat
Unit placebo (SCM)	Reassign "treatment" to each control unit	True effect should be an outlier
Location placebo (RDD)	Place cutoff at wrong value	No discontinuity should appear

SCM Permutation Test: The Classic

Abadie, Diamond & Hainmueller (2010): Run SCM for *every* control unit as if it were treated.

Compare post/pre RMSPE ratios:

- If the true treated unit's ratio is much larger than any placebo ratio → effect is robust
- The fraction of placebos with ratio ≥ true ratio = **p-value**



Source: Simulated data. Full replication code in `scripts/generate_lec09_placebo.py`.

Interpreting the Placebo Figure

Left panel: Gap in outcome over time.

- Gray = placebo gaps; Red = true treated gap
- Placebos hover near zero post-treatment; the true effect stands out

Right panel: Post/pre RMSPE ratio distribution.

- Red line = true unit's ratio (4.64)
- p-value ≈ 0.00 → none of 20 placebos produced a gap this large
- The SCM estimate is not driven by spurious pre-trend fitting

> ***Rule of thumb:*** *p-value < 0.05 (ratio in top 5%) → effect is genuine.*

Part 8

Extensions & Future Directions

Where the Literature Is Heading

Three active frontiers that extend the tools covered in this lecture:

DIRECTION	CORE IDEA	REPRESENTATIVE REFERENCES
DID + SC fusion	Make DID & SC doubly robust backups for each other	Sun, Xie & Zhang (2025), arXiv:2503.11375
Structural DID + ML	Embed DID logic within DML for HTE under high-dimensional confounding	Yu & Xu (2025), arXiv:2507.09718
Autocorrelation-robust inference	Valid SEs under serial corr. in long panels	Rambachan & Roth (2023); Conley (1999) spatial/clustered SEs

> *These are **extensions** — not core curriculum. The baseline toolkit remains CS2021 / SyDiD / DML-DiD / ML-RDD.*

Key Takeaway

> *"The latest direction braids together DID's identification logic, SC's weighting, ML's high-dimensional capacity, and staggered adoption's temporal complexity. Master the baseline first; these extensions layer on top."*

Further Reading: Classic References

PAPER	AUTHORS	VENUE
Difference-in-Differences with Variation in Treatment Timing	Goodman-Bacon (2021)	*JoE*
Difference-in-Differences with Multiple Time Periods	Callaway & Sant'Anna (2021)	*JoE*
Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program	Abadie, Diamond & Hainmueller (2010)	*JASA*
Synthetic Difference-in-Differences	Arkhangelsky et al. (2021)	*AER*

Further Reading: Methods & Recent Work

PAPER	AUTHORS	VENUE
Double/Debiased Machine Learning for Difference-in-Differences Models	Chang (2020)	*Econometrics Journal*
Flexible Covariate Adjustments in Regression Discontinuity Designs	Noack, Olma & Rothe (2024)	Working Paper
Difference-in-Differences Meets Synthetic Control: Doubly Robust Identification & Estimation	Sun, Xie & Zhang (2025)	arXiv
Bridging Structural Causal Inference & Machine Learning: The S-DIDML Estimator for Heterogeneous Treatment Effects	Yu & Xu (2025)	arXiv

Data & Code

This lecture:

- Slides source: `_slides/Lecture09-Difference-in-Differences-and-RDD-Slides.md`
- Placebo figure code: `scripts/generate_lec09_placebo.py`
- Exercises: `exercises/Lecture09-Exercise.ipynb`

Replication packages:

- Callaway & Sant'Anna (2021): `did` R package (`att_gt()`)
- Arkhangelsky et al. (2021): `synthdid` R package
- Chang (2020): [GitHub]

Thank You

Next: Lecture 10 — Optimal Policy Learning & Text as Data

Reading: Callaway & Sant'Anna (2021); Chang (2020)